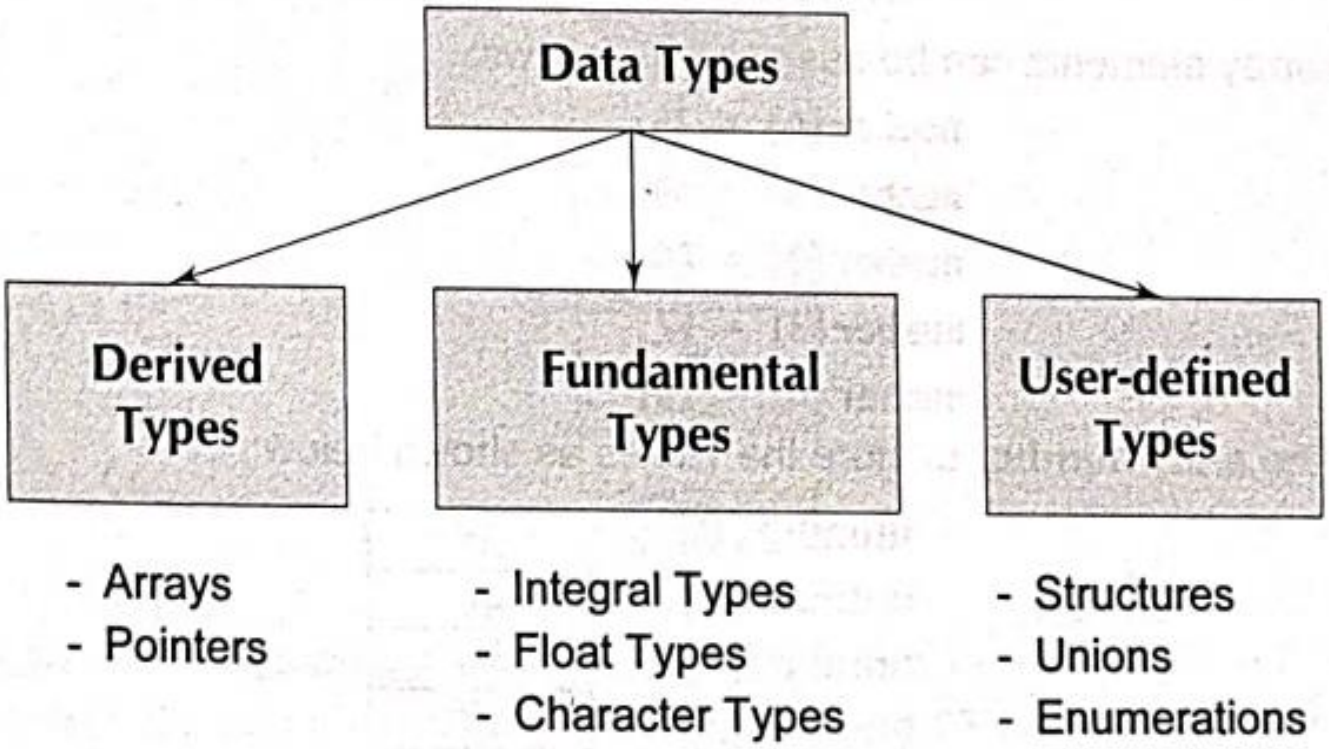
Chapter-7

Arrays

Data Types In C



Why?

- Consider the following Program,

```
#include <stdio.h>
 void main ()
{
   int x;
   x=5;
   x=10;
   printf("x=%d", x);
 }
```

What will be the value of x?

Cont…

- Variables are capable of holding only one value at a time.
- So far we have used only the fundamental data types, namely char, int, float, double and variations of int and double.
- Although these types are very useful, but a variable of these types can store only one value at any given time.
- So, they can be used only to handle limited amounts of data.
- In many applications, however, we need to handle a large volume of data in terms of reading, processing and printing.

Cont…

- However, there are a situation in which we want to store more than one value at a time in a single variable.

  - Suppose we wish to arrange the percentage marks obtained by 100 students in ascending order.

- We have two options to store these marks in memory.

a) Declare 100 variables to store percentage marks obtained by 100 different students, i.e. each variable containing one student's marks.

b) Construct array(subscripted variable) capable of storing all hundred values.

Examples where the concept of an array can be used.

- List of temperatures recorded every hour in a day, or a month, or a year.
- List of employees in an organization.
- List of products and their cost sold by a store.
- Test scores of a class of students.
- List of customers and their telephone numbers.
- Table of daily rainfall data.

Introduction to arrays

- An array is a fixed size sequenced collection of elements of the same data type and share a common name.

- **Types of Arrays**

- One Dimensional Arrays

- Two Dimensional Arrays

- Multidimensional Arrays

One-dimensional array

- A list of items can be given one variable name using only one subscript and such a variable is called a one-dimensional array.

Declaration of One-dimensional array,

General form

type variable-name [size];

- Type specifies the type of element that will be contained in the array.

ex. int, float, char etc.

Cont…

- size – is the maximum number of elements that can be stored inside the array.

- The subscript or index of an array can begin with number 0.

- That is x[0] is allowed. And refers to the 1st element of the array x[10].

- We can define an array name **per** to represent a set of percentage marks of a group of 100 students.

  i.e. **float per[100];**

Cont…

- Particular value is indicated by writing a number called index or subscript in brackets after the array name.

For example,

per[9]

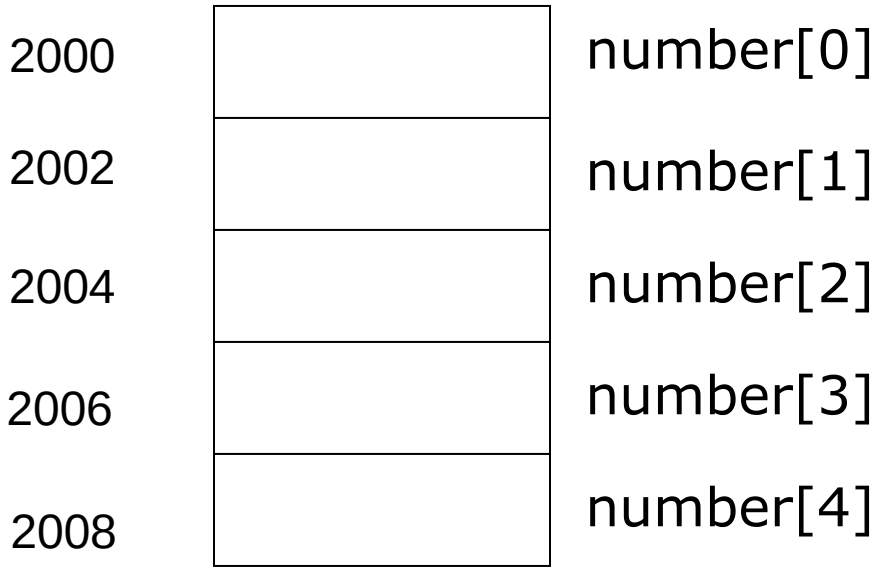
Represents the percentage marks of 10th student.

Example

- If we want to represent a set of five numbers with an array variable number then first we have to declare it as:

int number[5];

- The computer reserves five storage locations as,



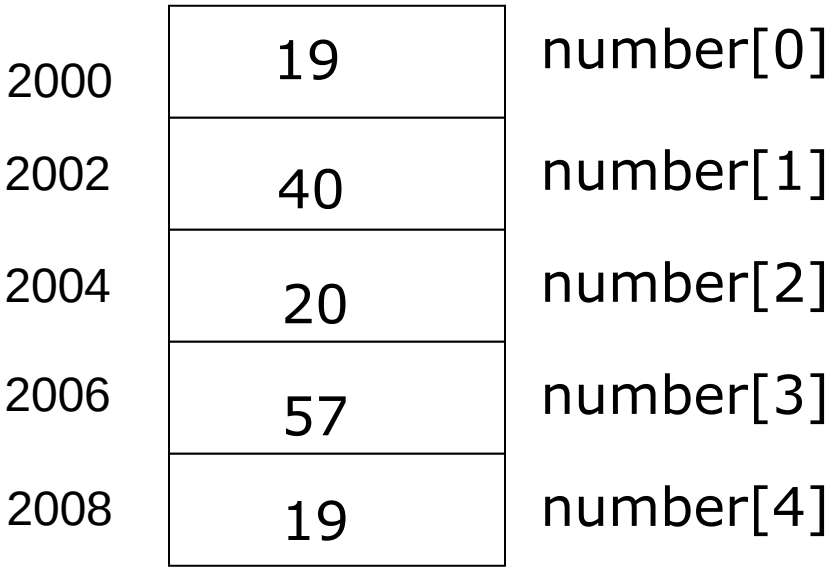
Cont…

- Value to the array elements can be assigned as:

```
number[0] = 19;
number[1] = 40;
number[2] = 20;
number[3] = 57;
number[4] = 19;
```

Cont…

- This would cause the array number to store the values as:



Cont…

- These elements may be used in programs just like any other C variable.

```
a = number[0] +10;
number[4]=number[1] + number[2];
```

- The index of an array can be integer constants or any expressions that results in integer.

Advantage of arrays

- The ability to use a single name to represent a collection of items.
- So items can be referred by specifying the item number.
- For example, a loop with the index of array as control variable can be used to

  - read the entire array.
  - perform calculation and print out the results.

Bound checking

- C performs no bound checking for array and therefore, care should be taken to ensure that the array index is within the declared limits.

- Because any reference to the arrays outside the declared limits would not necessarily cause an error. Rather, it might result in unpredictable program results.

Initialization of array

- **Compile time initialization**

➢ Initialize the elements of array when they are declared,

type arrayname[size]={list of values};

ex. int arr[5]={10,20,25,30,40};

int arr[5]={1,2,3};

then the remaining two elements will initialize to zero.

Cont…

- The size may be omitted. In such cases, compiler allocates enough space for all initialized elements

ex. **int counter[ ]={1,1,1,1};**

will declare the counter array to contain four elements with initial values 1.

- **int number [3] ={ 10,20,30,40};**

  Compiler will produce an error.

Run Time Initialization

int arr[100];

- First 50 to zero and last 50 to one.

```
for (i=0;i<50;i++)
  {
     arr[i] = 0;
   }
for (i=50;i<100;i++)
   {
     arr[i] = 1;
    }
```

Cont…

- We can also use a scanf to initialize an array,

```
int x[2];
scanf( “%d %d”,&x[0] ,&x[1] );
```

Program: Print array elements

```
void main()
{
    int a[5]={11,12,13,14,15};    //   {11,12,13};
    int i;
  /*printf("%d\n",a[0]);
    printf("%d\n",a[1]);
    printf("%d\n",a[2]);
    printf("%d\n",a[3]);    //0
    printf("%d",a[4]);  */  //0

    for(i=0;i<5;i++)
  {
     printf("%d\n",a[i]);
   }
}
```

Program : Read array elements

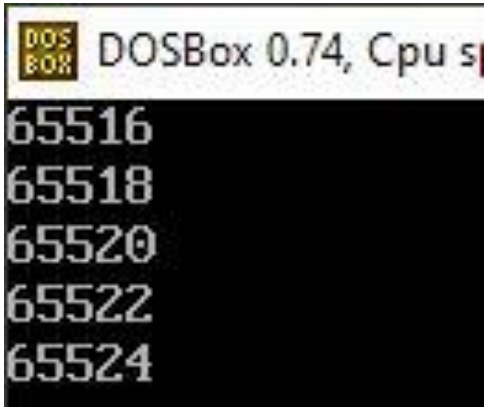
```
void main()
{
    int a[5];
    int i;
 /* scanf("%d",&a[0]);
    scanf("%d",&a[1]);
    scanf("%d",&a[2]);
    scanf("%d",&a[3]);
    scanf("%d",&a[4]);  */

    for(i=0;i<5;i++)      // for(i=0;i<4;i++)
  {
     scanf("%d",&a[i]);
   }
    //printf("%d",a[4]); //garbage value
}
```

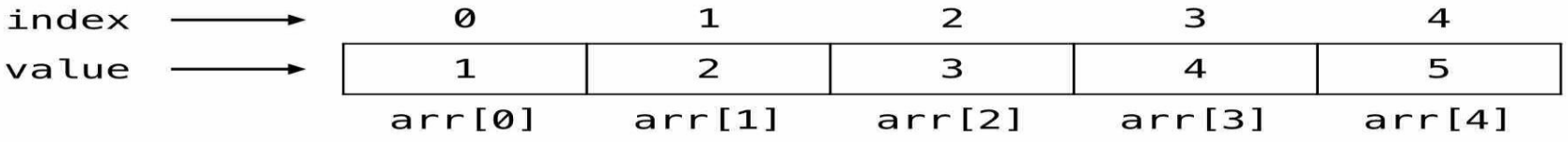
Program: Size of array & array element

```
void main()
{
    int a[5];
    printf(“%d”,sizeof(a));     //10 bytes
    printf(“%d”,sizeof(a[2])); //2 bytes
}
```

Program: Address of Array

```
void main()
{
    int arr[5]={1,2,3,4,5};
    int i;
    for(i=0;i<5;i++)
  {
     printf("%u\n",&arr[i]);
   }
}
```





Size of array

```
void main()
{
    int a[10];
 }
```

```
#define M 20
void main()
{
    int a[M];
 }
```

```
#define M 20
#define N 10
void main()
{
   int a[M+N*2];
 }
```

```
void main()
{
    int n;
    int a[n];
    scanf("%d",&n);
 }
//Not valid.
//Supported by gcc compiler
```

Program: Sum and Average

```
#include<stdio.h>
void main()
{
 int a[5],i;
 float avg;
 int sum=0;
 for(i=0;i<5;i++)
 {
  scanf("%d",&a[i]);
  }
 for(i=0;i<5;i++)
 {
  printf("%d\n",a[i]);
  }
```

```
for(i=0;i<5;i++)
{
  sum=sum+a[i];
 }
 avg=(float)sum/5;
 printf("sum=%d",sum);
 printf("avg=%f",avg);
}
```

Search Array Element.

```
#include<stdio.h>
void main()
{
   int a[10];
   int i,n,b;
   int pos,match=0;
   printf("enter number of elements:");
   scanf("%d",&n);
   for(i=0;i<n;i++)
   {
     scanf("%d",&a[i]);
   }
```

```
 printf("enter element to be search:");
 scanf("%d",&b);
 for(i=0;i<n;i++)
   {
      if(a[i]==b)
      {
         match=1;
         pos=i;
         printf("position=%d\n",i);
           // break;
       }
     }
if(match==0)
      {
         printf("no match found");
       }
}
```

swap max and min number

```
#include<stdio.h>
void main()
{
  int a[10],i,max,min;
  int pos1=0,pos2=0;
  int temp,n;

  scanf("%d",&n);

  for(i=0;i<n;i++)
  {
   scanf("%d",&a[i]);
  }
```

```
max=a[0];
min=a[0];

for(i=1;i<n;i++)
  {
     if(a[i]>max)
      {
         max=a[i];
         pos1=i;
      }
     if(a[i]<min)
     {
        min=a[i];
        pos2=i;
     }
   }
```

```
   printf("min=%d",min);
   printf("max=%d\n",max);

   temp=a[pos1];
   a[pos1]=a[pos2];
   a[pos2]=temp;

  for(i=0;i<n;i++)
  {
   printf("%d\n",a[i]);
  }
}
```

Insert Array Element.

```
#include<stdio.h>
void main()
{
   int a[10],i,n,b,pos;

   printf("enter number of elements:");
   scanf("%d",&n);

   for(i=0;i<n;i++)
   {
     scanf("%d",&a[i]);
    }

   printf("enter element to be insert:");
   scanf("%d",&b);
```

```
printf("position of element:");
scanf("%d",&pos);

for(i=n;i>pos;i--)
   {
      a[i]=a[i-1];
   }

   a[pos]=b;

   printf("new array after insertion is:\n");
   for(i=0;i<=n;i++)
   {
     printf("%d\n",a[i]);
    }
 }
```

More Programs……(In Lab Session)

- Delete Array Element.
- Sort Elements of Array.

Two-dimensional array

- Situations where a table of values will have to be stored.

Student	Mat	Phy	Chem.
Student # 1	89	77	84
Student # 2	98	89	80
Student # 3	75	70	82
Student # 4	60	75	80
Student # 5	84	80	75

Declaration of 2-D Array

- General form

  type arrayname[row_size][column_size];

  int stud[4][2];

- First subscript of the variable stu is row number which changes for every student.

- Second subscript contains two columns,
  - zeroth column which contains rollno
  - first column which contains marks

Cont…

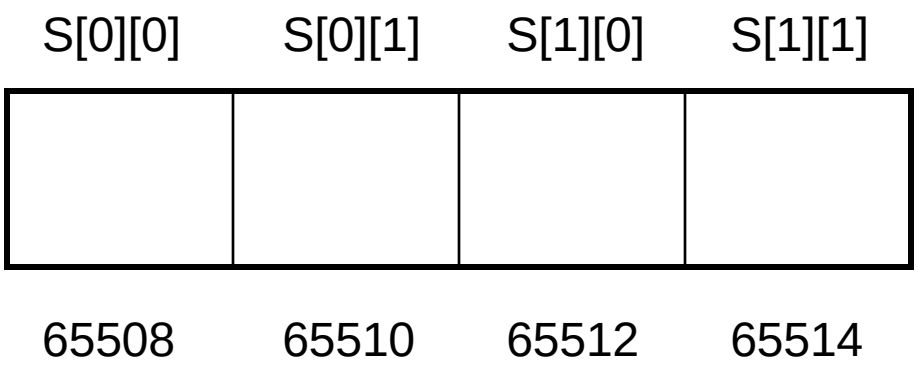
- Complete array arrangement is shown below.

	0	1
0	Stud[0][0]	Stud[0][1]
1	Stud[1][0]	Stud[1][1]
2	Stud[2][0]	Stud[2][1]
3	Stud[3][0]	Stud[3][1]

12	56
13	33
14	80
15	78

Memory map of 2-dimensional array

int s[2][2];



Initializing 2-dimensional array

- General format

type arrayname[row-size][col-size] = {list of values};

- int a[2][3] = { 0,0,0,1,1,1 };

initialize the elements of first row to 0 and second row to 1.

- The above statement can be equivalently written as

```
int a[2][3]= { { 0,0,0} , { 1,1,1} };
```

- We can also initialize a two dimensional array in the form of a matrix as,

```
int a[2][3]= {
                { 0,0,0} ,
                { 1,1,1}
             };
```

- When the array is **completely initialized with all values**, explicitly we need not specify the size of first dimension.

```
int a [ ] [3] = { { 0,0,0} , { 1,1,1} };
```

- When all elements are to be initialized to zero, shortcut method may be used,

```
int m [3] [5] = { { 0 } , { 0 } , { 0 } };     OR   ={0};
```

Run time:-

```
void main()
{
      int a[2][2];
      int i,j;
      for(i=0;i<2;i++)
      {
         for(j=0;j<2;j++)
        {
            scanf("%d",&a[i][j]);
         }
      }
 }
```

	0	1
0	a[0][0]	a[0][1]
1	a[1][0]	a[1][1]

1	2
3	4

Print Elements of 2D array

```
for(i=0;i<2;i++)
  {
        for(j=0;j<2;j++)
         {
            printf(“%d\t”,a[i][j]);
         }
    printf(“\n”);
   }
```

Thank You